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Target-following Robotic Platform Based on UWB Localization and Depth Camera

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Abstract: A mobile robot capable of following a specified target using Ultra-Wideband (UWB) technology and an RGB-D camera is presented in this paper. One of our primary objectives is to promote the use of UWB as a precise and fast real-time localization system (RTLS) in mobile robotics, specifically the target following applications using the approach of non-fixed anchors. We use the depth camera as a complementary sensor to the omnidirectional UWB, to increase the robot's awareness of its surroundings and possibly also as a way of providing human-machine interaction. This paper also describes creating a simplified mathematical model for computing the control action and the method we have used to measure the system's rotational dynamics for system identification purposes. Thereafter, we combined a PD controller and proportional virtual spring regulation and selected parameters of the compound regulator to slowly pursue a non-linear movement of the followed person. Our robotic platform is originally meant to carry a toolbox and equipment. However, it might be just as well used in various applications such as material handling in industrial and manufacturing settings, as an assistive technology for load handling in warehouses, construction sites, even carrying airline baggage or military equipment. Most countries implement regulations regarding lifting and carrying loads that are even more strict in case of repetitive or long-term use. Therefore, using a robotic platform in such cases might be convenient.

Keywords: Mobile robots; following robots; automatic guided vehicles and robotic vision.

1. INTRODUCTION

When it comes to robots carrying a load along a path, mobile robotics as a scientific field has numerous subcategories. Typically, when the path is predetermined, and we know the environment in advance, we use Automated-Guided Vehicles (AGV). A more modern variation of these devices is called Automated Mobile Robot or AMR. These are usually equipped with sensors and control logic that enables them to navigate through an unknown environment following dynamically moving goals. However, the difference between the two might be fuzzy, as is the case with the robot that we have developed.

Another important aspect of these robots is the degree to which they cooperate with humans. To enable human-robot interactions, often abbreviated as HRI, additional sensors are often included. To minimize redundancy and maximize cost-effectiveness, we believe such a sensor should be capable of combining the navigation functionality with HRI. The human following function is useful in many use cases such as assistance robots, robots used for equipment carrying, or heavy load manipulation, see Dedek et al. (2017); Ozana et al. (2014).

Regarding sensors, there is a group of refined and proven solutions that are often used with human following robots. Vision-based systems dominate the field, where LiDARs and cameras are often supplemented by proximity sensors aiding in collision detection. These solutions have already been successfully implemented and documented over time, such as in these publications Correa et al. (2012); Keat and Ming (2012). Researchers in Leigh et al. (2015) presented an especially interesting method, where they detect legs from 2D laser-scans. Vision-based systems are, as acknowledged by Deremetz et al. (2020); Feng et al. (2018), often sensitive to specific lighting conditions. Also, optical sensors such as cameras and LiDARs require line-of-sight (LOS) visibility. Wireless communication technologies are also common and often used for localization. This category includes, among others, GPS, Wi-Fi, BLE, or RFID. Where GPS lacks indoor precision, the others are more-or-less successfully used for non-line-of-sight (NLOS) human following navigation. It must be stated that even sensors in the wireless category have restrictions on various conditions, such as anchor placement, transmission speed through various materials, signal reflections, or frequency range of the selected technology.

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For our robot, we have combined both vision-based and wireless technologies to create a basic robotic target-following platform capable of indoor and outdoor navigation and obstacle detection. It is important to note that we are purposefully referring to our robot as a target following instead of a person or human following, as it is possible to place the measuring tag on any moving target. The platform shall be used as a basis for future academic works and thus it is substantial for us to make it modular and extensible. The resulting platform should model an assistive robot used for carrying tools and materials around a relatively open workplace with few obstacles.

2. RESEARCH OF RELATED TECHNOLOGIES

The distinctive technological feature of our robot is the use of ultra-wideband tags for localization. UWB is, strictly speaking, not a new technology. The first set of regulations allowing public use of this technology was released by US Federal Communication Commission in 2002. However, major organizations aiming to promote the widespread use of UWB, specifically UWB Alliance and FiRa Consortium, were founded rather recently in 2018 and 2019, respectively.

According to the aforementioned organizations and comparative research such as Gladysz and Santarek (2017), UWB compared to other RTLS technologies offers high precision down to 0.1 m, and resistance to signal reflection and interference.

To facilitate UWB measurements, we selected the Pozyx development kit. This kit includes hardware based on Decawave DW1000 UWB module, BNO-055 motion sensor, and proprietary positioning algorithms based on wireless two-way ranging, including an already implemented proprietary tracking algorithm. In addition, there are optional averaging settings, a possibility for setting UWB parameters, and documented API for Python.

There are publications available that have focused on researching applications of UWB in human following robots. Larger robots such as Deremetz et al. (2020); Ding et al. (2020) and Laneurit et al. (2016) are not well suited for confined indoor applications. On the other hand, very small solutions such as Ortiz et al. (2017) are bound to have problems with the geometric dilution of precision, if carrying its own anchors, reducing its following accuracy as shown in figure 1 limiting its use for applications with fixed anchors. We have found two prototypes similar to our desired solution. Human-following robot proposed by a Chinese team in Feng et al. (2018) is well documented, although lacking some functionalities that we wanted to implement, such as the use of RGB-D camera and heavy-duty, robust hardware design. There is also a prototype of a human-following robot built with a Pozyx kit ToddL (2018). Unfortunately, we could not find any additional information about this specific build as it seems to be a homemade hobby solution. Two previously mentioned solutions are demonstrating many technologies and algorithms that we later implemented in our robot.

In addition, we have implemented additional sensors and algorithms and a unique hardware design in our solution to further optimize the capabilities and usability of the resulting robotic platform. We could not find any references to previously published papers documenting the same combination of sensors and algorithms in similar applications.

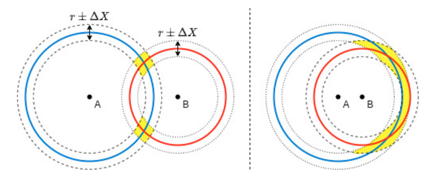

Fig. 1. Geometric dilution of precision in localization.

3. OBJECTIVES AND SCOPE

We have developed our platform while aiming to provide a practical, usable, and extendable solution. Even as a research prototype, it should already be usable as a load-carrying human-following assistance robot with a loading capacity of up to 200 kg. It stems from previous academic work at our university and is modular and open-sourced enough to be extensible and reusable as a platform for further academic research. Suggestions on subsequent topics to be further researched are listed in the section 8.

Our robot is combining two RTLS technologies, UWB and RGB-D, resulting in precise information about its user's location, while also enabling simple collision detection. Additional collision detection sensors are being added as part of subsequent development, to more robustly avoid collisions, especially during rotational movements. The intended purpose of the platform is heavy tools or toolbox transportation. Possibly it could be extended to function as a small-load forklift, and currently, we are planning to reuse obtained knowledge to advance our automatic parking facilities research. As there is no mapping or navigation implemented in the current version, "advanced AGV" using a human navigator as its guiding system seems more appropriate as a category label than "AMR", but might depend on the preferred definition. The usage of UWB as a precise and reliable RTLS in the short range, as well as a fusion of UWB measurement with other sensor data such as RGB/IR cameras or collision detection/avoidance detectors are key aspects of this project.

4. SYSTEM ARCHITECTURE

We developed the whole mechatronic solution for this robot, including protection circuits, an emergency-stop circuit, hardware chassis etc. For clarity of this paper, most of these fundamental steps are omitted.

One of these aspect, that deserves to be mentioned, is that a battery with a capacity of 17.4 Ah and a nominal voltage of 48 V is powering the system. This capacity is large enough to power the robot for prolonged periods of time with no issues. As shown in figure 2, devices are powered using different voltage levels and either communicate using USB or RS485, and in case of the IR sensor directly by 3.3–5V logic.

The robot is built using Jetson Nano single-board computer (SBC) as the main controller, allowing for onboard AI image processing and possessing a high-enough clock rate to facilitate the regulation. It is running Linux4Tegra

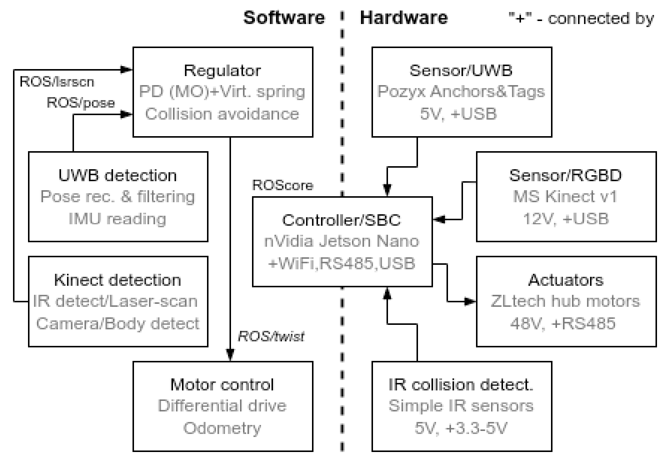

Fig. 2. Block diagram of the mechatronic solution

based on Ubuntu, which makes it possible to run ROS. It is important to mention that the regulation loop is currently limited not by Jetson's processing power, but by Pozyx reading frequency limitation divided by the number of anchors. Using the settings we selected based on the local regulations posed on UWB, the frequency lies between 15–20 Hz.

A PC can be connected to the robot remotely over Wi-Fi, leveraging ROS's distributed computing capabilities, which are especially useful for data monitoring and visualization. This also enables remote control when the internet connection is available.

The architecture of the system is based on publisher–subscriber model facilitated by ROS topics. Separate script nodes are reading the sensor's data and convert them into ROS messages, placing them into a callback queue. The controller script is reading these inputs and generates the control action. It publishes the results, that are subscribed by the motor-controlling code to convert them into speed commands for motors.

5. SOFTWARE SOLUTION

The software is built on top of the ROS framework. To ensure interoperability of used libraries, ROS and OS (Linux4Tegra) ROS 1 version Noetic was selected. 100 % of the code was written in Python 3. Functionality is added either as ROS nodes/servers or adapter Python classes for used libraries, such as Pozyx-Python-library.

Using ROS, it was also possible to fully simulate the robot in advance to test the differential drive, remote controller, depth-image to laser-scan transformation, and control algorithms. These tests were simulated using the Gazebo software. We have created a 3D model of the robot's chassis and created an URDF description using the model as its visual representation as seen in figure 3. A demo following application was then simulated using a Gazebo actor with a pre-programmed path and ROS server publishing its current location instead of the UWB measurement.

It was already hinted in the block diagram in figure 2, that the ROS nodes communicate with each other using ROS topics. The UWB detection node uses the Pozyx-Python-library to obtain measured distance data from the connected anchors. It is also validating the settings and reading the IMU measurements, that are also published through a topic. The Kinect measuring node is using the libfreenect library to read out the registered depth images, it later either converts them to laser-scans or track and detect poses/skeletons.

The code is separated into ROS nodes that loosely follow the mechatronic block diagram show in figure 2. Both the sensor and controller nodes are discussed later in respective sections 6 and 7. The drive node is using multiple Python classes to compute both the control action for both wheels and odometry from the encoder readings.

The motors related code is based on a Modbus interface for the drivers in Python and the following differential drive model. The linear component of the movement might be described as the average of both wheels' angular velocities proportional to their radius. Similarly, the angular component arises from the difference in the wheels' velocities proportional directly to their radius and inversely to their distance.

$$v = r\frac{(\omega_r + \omega_l)}{2} \tag{1}$$

$$\omega = \frac{r}{l}(\omega_r - \omega_l) \tag{2}$$

Representing these velocities mathematically, we can build equations 1 and 2. These equations are important for the regulator as they are used for decoupling. That is by combining these, it is possible to find the angular velocity of each wheel to use as a control action, as shown in equation 3.

$$\begin{aligned} \omega_r &= \frac{2v + \omega_l}{2r} \\ \omega_l &= \frac{2v - \omega_l}{2r} \end{aligned} \tag{3}$$

It is also possible to formulate the mathematical representation of the differential drive as a state-space model where the system is represented by three state variables $\dot{x}$, $\dot{y}$ and $\dot{\phi}$. Such a model described by equation 4 is useful for control system design in the section 7.

$$\begin{aligned} \dot{x} &= r\frac{(\omega_r + \omega_l)}{2}\cos(\phi) \\ \dot{y} &= r\frac{(\omega_r + \omega_l)}{2}\sin(\phi) \\ \dot{\phi} &= \frac{r}{l}(\omega_r - \omega_l) \end{aligned} \tag{4}$$

6. SENSORS AND MEASUREMENTS

The UWB is a fast and precise technology with undisputable advantages over real-time localization solutions. It is, however, not yet as widely used as the other typical RTLS technologies, such as RFID or LiDARS. Also, UWB is often used to localize a tag attached to an object in a previously known environment, where localization anchors have been installed. Our robot takes advantage of UWB's outstanding precision in localization that enabled us to sacrifice some of the precision and choose the approach of placing the anchors directly on the robot to substantially extend its operating range. UWB is often compared to a vision-based system to demonstrate its robustness in different environments and suitability for outdoor use. We have opted for a combination of the two as they complement each other effectively. UWB offers robust and omnidirectional localization, whereas a vision-based system can be advantageous in an unknown environment, improve precision near the operating point and assist in human-machine interaction.

In our localization system, the dominant role is the one of the UWB. Specifically, the Pozyx RTLS solution, that facilitates the UWB measurements and communications as well as filtering and IMU measurements. UWB channel and gain are set to meet the local regulations in the Czech Republic. Pozyx's "FAST" ranging protocol and "TRACKING" proprietary algorithm is used. To increase reading frequency, a high bitrate and short preamble length was selected.

As mentioned above, we have placed the anchors on the follower instead of spreading them in the environment. Such an approach is bound to reduce the measurement precision, as it violates multiple recommended rules of the anchor setup. The anchors should be placed as far as possible from metals, as it might alter the antenna's properties unfavorably. Also, the anchors should be placed as far as possible from one another and never in a straight line. This rule follows from the principle of geometrical dilution of precision, which is illustrated in figure 1. If the anchors that are measuring the distance are placed close to each other or in line, the measurement error is amplified as the error boundaries intersect over wider area.

To place the anchors as far as possible, we have compromised between the robots' dimensions and practicality and opted for a size of roughly standard half-sized EUR/EPAL pallet, that is $\approx 80 \times 60$cm. Therefore, it has a standardized size for transporting various industrial-grade crates, boxes, box pallets, etc. while still being reasonably compact.

To test that the measurement is precise enough, even after violating the aforementioned rules, an experimental measurement with visualization was conducted. A best case, where the tag is one meter apart from the robot (which is the minimal following distance), aimed directly at the robot and completely static is shown in figure 3.

It is important to mention that such precision is only possible in favourable conditions such as with the tag pointing straight towards the robot and providing enough time for the measurement to settle. The latter, is especially important because the filtering set in the tracking algorithm creates a substantial delay. Understandably, in less favourable conditions, the precision deteriorates.

The obtained reading indicates $\approx 0.07m$ of difference, which is sub-decimeter precision. Meaning that even if the reading precision degrades by an order of magnitude, it is still viable for use in human tracking, as an error of up to a meter is generally acceptable. Also, thanks to this level of precision, in practice, the following function might be gracefully terminated simply by placing the tag on the robot.

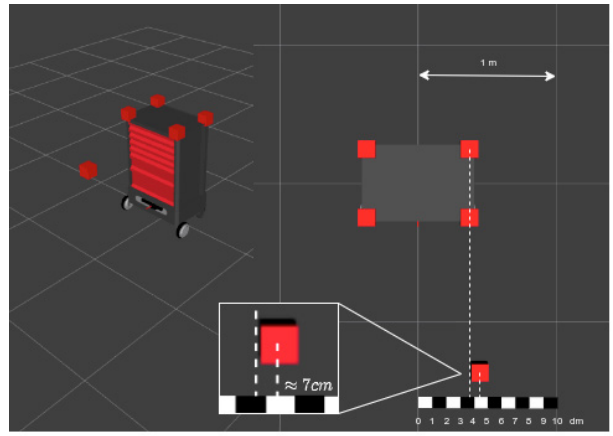

Fig. 3. Precision measurement.

The position and orientation obtained by Pozyx are then transformed into the ROS-specific type "Pose", which comprises coordinates and quaternion. We might then use the measured Pose to calculate the regulation error.

The issue of a measurement delay can be mitigated by the use of the complementary sensor. If the leg_tracker or skeleton centroid data are available, these could be preferred as they are more precise and have low latency. On the other hand, visual data are not omnidirectional and due to Kinect's limited field of view, it is very easy to lose the user from a line-of-sight (LOS) view. That is why for the general following, we prefer the UWB Pose measurements.

Regarding the vision-based sensors, we have selected RGB-D cameras. They offer a wide range of uses, contributing to the extendability of the robot's functionality. The possible uses have already been stated above.

We opted for the Kinect v1 sensor to use as the RGB-D camera, as it is a cheap option with well-documented libraries offering both color and IR readings. It was used in combination with the OpenKinect/libfreenect library and freenect_stack ROS package, so the final ROS formatted outputs might easily be replaced by more advanced sensors later. There are many papers describing the use and advantages of this sensor in depth, such as Keat and Ming (2012) or Correa et al. (2012).

A fast method of using depth-image obtained by RGB-D camera is to convert it to 2D laser-scan. This transformation discards a large portion of useful data, reducing the 2D depth image to a 1D depth line. The advantage of this approach is the ability to process such datasets very fast. We have successfully used the AI detector described in the following article Leigh et al. (2015), to scan and detect legs in the scan with little modification.

Another possible use of the laser-scan, is to provide additional data to the obstacle avoidance system. As there is no mapping or navigation implemented at the moment, the avoidance algorithm is purely reactive as it detects there is an obstacle ahead. It can either simply halt, or force a 90-degree change in a random direction. However, the latter method would also react to the followed target, thus forcing an undesired turn, while approaching the target.

We have implemented simple collision detection using laser-scans by dividing the range into angular sections and checking for obstacles. Even though this approach worked in Gazebo simulation, we didn't deploy this on the real robot, dismissed due to the limited field of view of the Kinect sensor.

An alternative approach is using the built-in Maxwell GPU of Jetson Nano to process the image and find body parts/skeleton. For skeleton detection, there is a trt_pose AI library available specifically for Jetson jaybdub (2023). It is elementary to compute, e.g. a centroid of this skeleton body to get the user's position. The advantage of this approach is that even a semi-hidden body might be tracked as seen in figure 4. The main disadvantage is a substantially higher GPU and memory load. We have implemented this optional functionality as a proof of concept for reference. Also, there is a gesture detector trt_pose_hand that might be used, e.g. as a simple hardware-less HRI interface. However, that functionality is yet to be implemented.

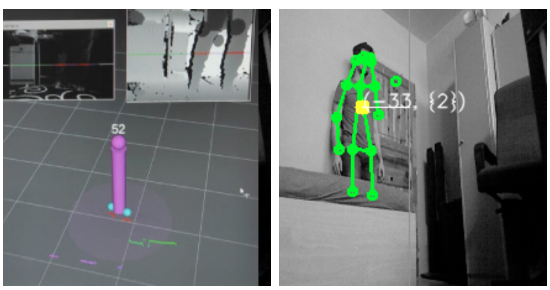

Fig. 4. Demonstration of the leg detector Leigh et al. (2015) and skeleton tracking with an obstacle.

7. CONTROL SYSTEM

As is shown in figure 5, there are two regulators present in the control structure, both have their own fixed set point, and actuating signal. To avoid oscillation caused by measurement errors, hysteresis around the setpoints was set. e.g. distances in sufficiently close vicinity of the set point are considered equal to the setpoint.

A method for identification of the system rotational dynamics was derived to enable the control design, see figure 6. By setting angular velocity ω to a constant value while measuring the step function of angular position ϕ from motor encoders, we can get the 2nd-order astatic system characteristic. To allow further analysis, we had to differentiate the angular position, to obtain back the angular velocity ω. Using this workaround, we got 2nd-order static dynamical characteristic that is easier to process. Finally, we must not forget to multiply the identified system in the Laplace transform by an integrator, to nullify the effects of the derivation. The experimentally achieved transmission of the system is given in equation 5.

$$G(s) = \frac{1.01}{(1+0.02625s)(1+0.1146s)}\frac{1}{s} \tag{5}$$

For the angle deviation regulator, we opted for the standard PD regulator. Initially we tried finding the parameters using modulus optimum. Equation 5 describes the closed-loop frequency transfer function. We can find a modulus of this transfer function. The result was regulator with the parameters given in the following equation:

$$\begin{aligned} G_R(s) &= 19.664 + 2.275s \\ G_W(s) &= \frac{1}{(1+0.02558s)(1+0.2561s+0.182s^2)} \end{aligned} \tag{6}$$

For the distance deviation regulator, we used the aforementioned virtual spring regulation. A simple linearly proportional regulator was parameterized manually to behave as a virtual spring between the target and the follower. The further the target is the greater is the virtual force pulling the follower. On the other hand, if the follower gets too close, the virtual spring's force pushes in the opposite direction to return to the set-point distance.

There are several limiting factors affecting the regulation. First the parameters set to the UWB increased the measurement reliability and precision on one hand, but introduced larger delays on the other. Enabling filtering to crop out outlier values, introduced by the geometric dilution of precision further prolongs these delays. This means, there is a significant time lag up to around 2 seconds in worst case. Also the unpredictive nature of the human movement coupled with it's non-linearity in both direction and speed further complicates the controller's design. For the previously mentioned reasons, we found that scaling the P component of the rotational regulator to higher values, introducing bigger overshoot and mitigating it by adding hysteresis around zero is preferable to improve the following behavior. More precise regulation to rotate the robot towards the user can be achieved by switching the feedback reading to RGB-D camera while in the deadzone of the rotational regulator's hysteresis and rotating at lower speed to fine-tune the final rotation towards the target.

8. CONCLUSION

At the current development stage, the robot has been assembled as described in the text above as a successful proof of concept. It was built as a prototype and put into operation for testing, see figure 7. The target-following functionality was also tested in the university campus. To start the following, the user simply has to enable the motors via a provided button and attach the tag to the followed target or himself if he is to be followed (i.e. put the tag in a pocket).

The selection of ROS as the base framework should be helpful in this case, as the navigation stack might be a straightforward way to implement such functionality. Adding advanced navigation capabilities would greatly enhance the control possibilities and obstacle avoidance, as trajectory planning and more modern control approaches could also be engaged.

Regarding the precision of the UWB, as stated in the text the average precision is usually between 0.1–1 m. Repetitive measurements tend to contain error against the true value and are prone generate outlying false measurements when no filtering is applied. However, even when the conditions are not strictly met, the resulting precision is usually sufficient. NLOS measurements are

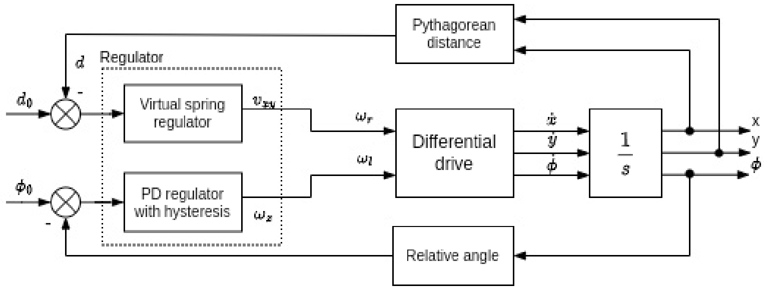

Fig. 5. Simplified control diagram.

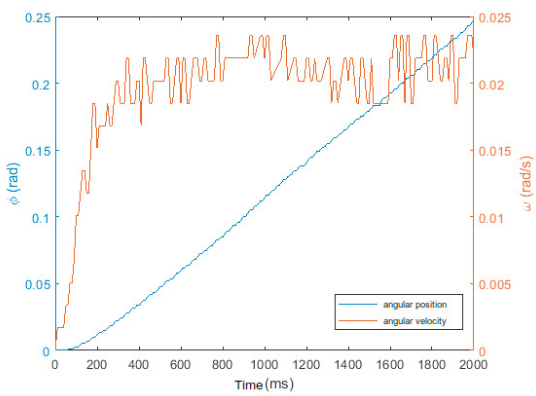

Fig. 6. Measured identification data compared to the theoretical idealised outline.


Fig. 7. Human following platform without load.

sometimes possible, the signal can transfer even through walls, if the distance is small enough (usually a first few meters after walking behind an obstacle). However, more sophisticated path planning would be required for the robot to be able to keep following in such scenario.

Follow-up academic work has already been started, to further develop this platform. Mainly to explore using more modern sensors to improve the collision detection. To upgrade and refactor the software stack to use ROS2. And also to improve upon the regulation algorithms described above by fusing data from the available sensors to achieve even better following behavior.

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